Part I: the Laplacian and the modularity matrix under unit degree

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Part I of *Modularity Matrix Approach for Graph Learning?*

Code in `modularity_graph_learning/`; every number below is printed by `modularity_graph_learning/report_numbers.py`.

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1 Summary

Under the paper's degree constraint $\mathrm{diag}(\mathbf{L}) = \mathbf{1}$, the modularity matrix is a reflection of the Laplacian, $\mathbf{B} = \mathbf{I} - \mathbf{L} - \mathbf{1}\mathbf{1}^\top/p$. Everything modularity could say about a learned graph is then already in $\mathbf{L}$:

1. **The lemma holds as stated**, with proofs below (Section 3). For (b) I give the exact form without the condition $\lambda_k \le 1$: the two sides differ by $\max\{0, \lambda_k - 1\}$. Two unit-degree counterexamples show the condition is needed (Section 4).
2. **The match is a statement about regular graphs, and more generally about normalised operators** (Section 5). With unequal degrees, the combinatorial $\mathbf{L}$ and $\mathbf{B}$ stop sharing eigenvectors.
3. **On the repaired Fig. 2 graph the prediction holds to within the solver's degree tolerance** (Section 6). k-means on the top two eigenvectors of $\mathbf{B}$ returns the three components exactly, and they are the three sectors.
4. **Loosening the degree penalty breaks the algebra but not the partition** (Section 7). The normalised identity keeps holding exactly throughout.

So, inside the paper's framework, modularity adds nothing new. Anything new would have to come from the choice of null model (Part II) or from dropping the degree constraint (Part III).

2 Setting

Let $\mathbf{W}$ be a $p \times p$ symmetric weight matrix with $W_{ij} \ge 0$ and $W_{ii} = 0$. Write $\mathbf{d} = \mathbf{W}\mathbf{1}$ for the degrees, $\mathbf{D} = \mathrm{Diag}(\mathbf{d})$, $2m = \mathbf{1}^\top\mathbf{d}$, the Laplacian $\mathbf{L} = \mathbf{D} - \mathbf{W}$ and the modularity matrix $\mathbf{B} = \mathbf{W} - \mathbf{d}\mathbf{d}^\top/2m$. Both satisfy $\mathbf{L}\mathbf{1} = \mathbf{B}\mathbf{1} = \mathbf{0}$ (for $\mathbf{B}$: $\mathbf{B}\mathbf{1} = \mathbf{d} - \mathbf{d}(\mathbf{1}^\top\mathbf{d})/2m = \mathbf{0}$).

Since $\mathrm{diag}(\mathbf{L}) = \mathbf{d}$, the constraint $\mathrm{diag}(\mathbf{L}) = \mathbf{1}$ is exactly $\mathbf{d} = \mathbf{1}$. Then $\mathbf{D} = \mathbf{I}$, $2m = p$, $\mathbf{W}$ is doubly stochastic, and

$$\mathbf{B} = \mathbf{W} - \tfrac{1}{p}\mathbf{1}\mathbf{1}^\top = \mathbf{I} - \mathbf{L} - \tfrac{1}{p}\mathbf{1}\mathbf{1}^\top. \tag{1}$$

This is a property of the *learned* graph, not an assumption about the data: every graph that Problems (6), (8) and (10) of the paper return satisfies it. The eigenvalues of a symmetric doubly stochastic $\mathbf{W}$ lie in $[-1, 1]$ (each row of $|\mathbf{W}|$ sums to one), so those of $\mathbf{L} = \mathbf{I} - \mathbf{W}$ lie in $[0, 2]$. Write them $0 = \lambda_1 \le \lambda_2 \le \cdots \le \lambda_p \le 2$ and those of $\mathbf{B}$ as $\mu_1 \ge \cdots \ge \mu_p$.

3 The lemma

Lemma 1 (Reflection)**.** *Under $\mathbf{d} = \mathbf{1}$, choose an orthonormal eigenbasis of $\mathbf{L}$ that contains $\mathbf{1}/\sqrt{p}$, so $\mathbf{L}\mathbf{v}_i = \lambda_i\mathbf{v}_i$ with $\mathbf{v}_1 = \mathbf{1}/\sqrt{p}$ and $\mathbf{v}_i \perp \mathbf{1}$ for $i \ge 2$. Then $\mathbf{B}\mathbf{v}_1 = \mathbf{0}$ and $\mathbf{B}\mathbf{v}_i = (1 - \lambda_i)\mathbf{v}_i$ for $i \ge 2$. Hence $\mathbf{L}$ and $\mathbf{B}$ share eigenvectors, and as multisets* $\mathrm{spec}(\mathbf{B}) = \{0\} \cup \{1 - \lambda_i : \ i = 2, \ldots, p\}$.

Proof. Such a basis exists because $\mathbf{1}$ lies in the kernel of $\mathbf{L}$. By (1), $\mathbf{B}\mathbf{v}_1 = \mathbf{v}_1 - \mathbf{L}\mathbf{v}_1 - \mathbf{1}(\mathbf{1}^\top\mathbf{v}_1)/p = \mathbf{v}_1 - \mathbf{0} - \mathbf{v}_1 = \mathbf{0}$. For $i \ge 2$, $\mathbf{1}^\top\mathbf{v}_i = 0$, so $\mathbf{B}\mathbf{v}_i = \mathbf{v}_i - \lambda_i\mathbf{v}_i$. □

So on $\mathbf{1}^\perp$ the spectrum of $\mathbf{B}$ is that of $\mathbf{L}$ reflected through $\frac{1}{2}$: small Laplacian eigenvalues (cheap cuts) become large modularity eigenvalues (good splits). The direction $\mathbf{1}$, which both matrices ignore, maps to 0 in both.

Proposition 1. *Under* $\mathbf{d} = \mathbf{1}$*:*

(a) $\operatorname{rank}(\mathbf{L}) = p - k$ *if and only if* $\mathbf{B}$ *has eigenvalue* 1 *with multiplicity exactly* $k - 1$*. Moreover* $\mu_1 \leq 1$*.*

(b) For every $k \geq 2$*,*

$$\min_{\mathbf{V}^\top \mathbf{V} = \mathbf{I}_k} \operatorname{tr}(\mathbf{V}^\top \mathbf{L} \mathbf{V}) \;=\; (k-1) - \max_{\mathbf{U}^\top \mathbf{U} = \mathbf{I}_{k-1}} \operatorname{tr}(\mathbf{U}^\top \mathbf{B} \mathbf{U}) \;+\; \max\{0, \lambda_k - 1\}.$$

In particular, if $\lambda_k \leq 1$ *the last term vanishes, and the Fan penalty of the paper's Algorithm 1 is, up to the constant* $k - 1$*, minus a relaxed* $(k-1)$*-way modularity.*

(c) $\mu_1 = \max\{0, 1 - \lambda_2\}$*.*

Proof. (a) For nonnegative weights, $\operatorname{rank}(\mathbf{L}) = p - k$ means $\mathbf{L}$ has eigenvalue 0 with multiplicity k, that is, exactly $k - 1$ of $\lambda_2, \ldots, \lambda_p$ are zero. By Lemma 1 these, and only these, give eigenvalue $1 - 0 = 1$ of $\mathbf{B}$; the remaining eigenvalue of $\mathbf{B}$, on $\mathbf{v}_1$, is $0 \neq 1$. Since every $\lambda_i \geq 0$, every $1 - \lambda_i \leq 1$, so $\mu_1 \leq 1$.

(b) By Ky Fan's theorem the left side is $\sum_{i \leq k} \lambda_i$, and the maximum on the right is the sum of the $k - 1$ largest elements of the multiset $\{0\} \cup \{x_i\}_{i \geq 2}$ with $x_i = 1 - \lambda_i$, where $x_2 \geq x_3 \geq \cdots$. If $x_k \geq 0$, those $k - 1$ elements are $x_2, \ldots, x_k$ (a tie with the 0 does not change the sum). If $x_k < 0$, then $0 > x_k$, and at most $k - 2$ elements exceed 0, so the 0 is among the top $k - 1$ and displaces x_k. In both cases the sum is $\sum_{i=2}^{k} x_i + \max\{0, -x_k\} = (k-1) - \sum_{i \leq k} \lambda_i + \max\{0, \lambda_k - 1\}$ (using $\lambda_1 = 0$). Rearranging gives the identity.

(c) By Lemma 1, the largest element of $\{0\} \cup \{1 - \lambda_i\}_{i \geq 2}$ is $\max\{0, 1 - \lambda_2\}$. □

The optimisers in (b) correspond too: when $\lambda_k \leq 1$, $\mathbf{V} = [\mathbf{1}/\sqrt{p},\ \mathbf{U}]$ with $\mathbf{U} = [\mathbf{v}_2, \ldots, \mathbf{v}_k]$. When the rank constraint of Algorithm 1 is fully active, $\lambda_1 = \cdots = \lambda_k = 0$, so $\lambda_k \leq 1$ holds automatically. The condition matters for iterates and approximate solutions.

Remark. In (c), $\mu_1 > 0$ is Newman's test for whether a network is divisible at all. So on a unit-degree graph the paper's gate $\lambda_2 < \tau = 1$ would be exactly that test. The same identity also discards everything above $\lambda_2 = 1$, since μ_1 is then 0.

4 The role of $\lambda_k \leq 1$

For a unit vector $\mathbf{v} \perp \mathbf{1}$, (1) gives $\mathbf{v}^\top \mathbf{B} \mathbf{v} = 1 - \mathbf{v}^\top \mathbf{L} \mathbf{v}$. So $1 - \lambda_i$ is the relaxed modularity of splitting the graph along eigenvector i, while not splitting (the direction $\mathbf{1}$) scores 0. $\lambda_i < 1$ means that split beats not splitting; $\lambda_i > 1$ means it is worse than doing nothing. The condition $\lambda_k \leq 1$ therefore says that there are at least $k - 1$ directions along which splitting is worthwhile. When it fails, the Fan term still asks for k groups while modularity prefers fewer, and Proposition 1(b) measures the disagreement: exactly $\lambda_k - 1$.

Two unit-degree graphs show the failure (Table 1). The complete graph K_4 with weights 1/3 has $\lambda_2 = 4/3 > 1$, so it fails already at $k = 2$. The 6-cycle with weights 1/2 has $\operatorname{spec}(\mathbf{L}) = \{0,\ 0.5,\ 0.5,\ 1.5,\ 1.5,\ 2\}$ and $\operatorname{spec}(\mathbf{B}) = \{0.5,\ 0.5,\ 0,\ -0.5,\ -0.5,\ -1\}$; the identity holds for $k = 2, 3$ and fails at $k = 4$, where $\lambda_4 = 1.5$.

5 When do L and B share eigenvectors?

In general, $\mathbf{B} = \mathbf{W} - \mathbf{d}\mathbf{d}^\top / 2m = \mathbf{D} - \mathbf{L} - \mathbf{d}\mathbf{d}^\top / 2m$. Each matrix has a direction it treats as uninformative: $\mathbf{L}$ ignores $\mathbf{1}$ (every node moving together), and the null term of $\mathbf{B}$ points along $\mathbf{d}$

graph	k	λ_k	Fan side $\sum_{i\le k}\lambda_i$	modularity side $(k-1)-\sum_{i<k}\mu_i$
K_4, weights 1/3	2	1.333	1.333	1.000
6-cycle, weights 1/2	2	0.5	0.5	0.5
	3	0.5	1.0	1.0
	4	1.5	2.5	2.0

Table 1: Proposition 1(b) on two unit-degree graphs. Where $\lambda_k > 1$ the sides differ by exactly $\lambda_k - 1$.

(nodes moving in proportion to degree). The two matrices share eigenvectors when the diagonal part is a multiple of the identity and the two uninformative directions agree.

setting	ignored by $\mathbf{L}$	ignored by $\mathbf{B}$	shared eigenvectors?
regular, $\mathbf{D}=c\mathbf{I}$ (includes $\mathbf{d}=\mathbf{1}$)	$\mathbf{1}$	$\mathbf{1}$	yes: $\mathbf{B}=c\mathbf{I}-\mathbf{L}-\frac{c}{p}\mathbf{1}\mathbf{1}^\top$
normalised, any degrees	$\sqrt{\mathbf{d}}$	$\sqrt{\mathbf{d}}$	yes: $\mathbf{B}_n=\mathbf{I}-\mathbf{L}_{\text{sym}}-\mathbf{u}\mathbf{u}^\top$
combinatorial, unequal degrees	$\mathbf{1}$	$\mathbf{d}$	no

Table 2: Three settings. $\mathbf{B}_n=\mathbf{D}^{-1/2}\mathbf{B}\mathbf{D}^{-1/2}$, $\mathbf{L}_{\text{sym}}=\mathbf{I}-\mathbf{D}^{-1/2}\mathbf{W}\mathbf{D}^{-1/2}$ and $\mathbf{u}=\sqrt{\mathbf{d}/2m}$.

Regular. If $\mathbf{d}=c\mathbf{1}$ then $2m=cp$ and $\mathbf{d}\mathbf{d}^\top/2m=c^2\mathbf{1}\mathbf{1}^\top/(cp)=\frac{c}{p}\mathbf{1}\mathbf{1}^\top$, so $\mathbf{B}=c\mathbf{I}-\mathbf{L}-\frac{c}{p}\mathbf{1}\mathbf{1}^\top$, and the proof of Lemma 1 goes through with 1 replaced by c: $\text{spec}(\mathbf{B})=\{0\}\cup\{c-\lambda_i\}$.

Normalised. For any positive degrees, since $\mathbf{D}^{-1/2}\mathbf{d}=\sqrt{\mathbf{d}}$,

$$\mathbf{D}^{-1/2}\mathbf{B}\mathbf{D}^{-1/2}=\mathbf{D}^{-1/2}\mathbf{W}\mathbf{D}^{-1/2}-\frac{\mathbf{D}^{-1/2}\mathbf{d}\mathbf{d}^\top\mathbf{D}^{-1/2}}{2m}=(\mathbf{I}-\mathbf{L}_{\text{sym}})-\mathbf{u}\mathbf{u}^\top.$$

Here $\mathbf{u}$ is a unit vector and $\mathbf{L}_{\text{sym}}\mathbf{u}\propto\mathbf{D}^{-1/2}\mathbf{L}\mathbf{1}=\mathbf{0}$, so the argument of Lemma 1 applies verbatim with $\mathbf{u}$ in place of $\mathbf{1}/\sqrt{p}$. This is the identity of the 8 September report, and Part I is its special case $\mathbf{d}=\mathbf{1}$, where $\mathbf{u}=\mathbf{1}/\sqrt{p}$ and $\mathbf{L}_{\text{sym}}=\mathbf{L}$. In particular, part (a) holds for the normalised pair *for any degrees*: $\mathbf{L}_{\text{sym}}$ has k zero eigenvalues exactly when $\mathbf{B}_n$ has eigenvalue 1 with multiplicity $k-1$.

Unequal degrees. For an eigenvector $\mathbf{v}$ of $\mathbf{L}$ with eigenvalue λ, $\mathbf{B}\mathbf{v}=\mathbf{D}\mathbf{v}-\lambda\mathbf{v}-\mathbf{d}(\mathbf{d}^\top\mathbf{v})/2m$. This is a multiple of $\mathbf{v}$ only if $\mathbf{D}\mathbf{v}-\mathbf{d}(\mathbf{d}^\top\mathbf{v})/2m$ is, which holds when $\mathbf{D}=c\mathbf{I}$ but fails in general. On a triangle with two tails, degrees $(2,2,4,2,1,1)$, the Fiedler vector $\mathbf{v}$ of $\mathbf{L}$ has $\|\mathbf{B}\mathbf{v}-(\mathbf{v}^\top\mathbf{B}\mathbf{v})\mathbf{v}\|=0.50$, so it is not an eigenvector of $\mathbf{B}$. The normalised pair matches to 4.0×10^{-16}: $\text{spec}(\mathbf{L}_{\text{sym}})=\{0,\,0.340,\,0.784,\,1.5,\,1.5,\,1.876\}$ and $\text{spec}(\mathbf{B}_n)=\{0.660,\,0.216,\,0,\,-0.5,\,-0.5,\,-0.876\}$. On a graph this small the sign split of the leading vectors still gives the same partition, so different eigenvectors do not automatically mean different clusters.

Normalisation is not a free fix

Normalising changes coordinates, not the graph, but it does change the statistical model. The paper's likelihood uses $\mathbf{L}$ itself as a precision matrix, $p(\mathbf{x})\propto\exp(-\frac{1}{2}\mathbf{x}^\top\mathbf{L}\mathbf{x})$; using $\mathbf{L}_{\text{sym}}$ there would be a different model. Inside the paper's framework, then, the only place modularity can carry information that $\mathbf{L}$ does not is the combinatorial pair with unequal degrees, that is, after dropping $\text{diag}(\mathbf{L})=\mathbf{1}$ (Part III).

6 Numerical check on the repaired Fig. 2 graph

The graph is our Algorithm 1 estimate on the paper's S&P 500 panel ($p=97$ stocks in three sectors, 753 daily returns from 5 January 2016 to 31 December 2018; $k=3$, $\eta=300$, degree penalty $\rho=100$), started from the Problem (1) estimate as described in the 18 October note. Its three components have 52, 28 and 17 stocks and are exactly the Industrials, Consumer Staples and Energy sectors (ARI 1.000).

Our solver enforces $\mathrm{diag}(\mathbf{L}) = \mathbf{1}$ by the quadratic penalty $\frac{\rho}{2}\|\mathrm{diag}(\mathbf{L}) - \mathbf{1}\|^2$, not exactly, so the degrees are close to one but not equal to it. Table 3 compares each prediction of Proposition 1 with what the learned graph gives.

quantity	Part I predicts	learned graph
degrees d_i	all 1	in [1.0012, 1.0069], mean 1.0047, CV 0.0014
$\|\mathbf{B} - (\mathbf{I} - \mathbf{L} - \mathbf{1}\mathbf{1}^\top/p)\|_F$	0	0.0479 (1.7% of $\|\mathbf{B}\|_F$)
three smallest $\lambda_i(\mathbf{L})$	0, 0, 0	all $\|\lambda_i\| \le 1.3 \times 10^{-15}$; next 0.3779
largest $\mu_i(\mathbf{B})$, (a)	1, 1, then $1 - \lambda_4$	1.0052, 1.0041, then 0.6274
Fan identity (b): $\sum_{i\le 3} \lambda_i$ vs $2 - \mu_1 - \mu_2$	equal	6.7×10^{-16} vs -0.0093
$\max_i \|\lambda_i - (1 - \mu_i)\|$ on $\mathbf{1}^\perp$ (96 pairs)	0	0.0059 (mean 0.0047)
same, reflecting about $c = \bar{d}$ instead of 1	0	0.0020 (mean 0.0006)
sin of largest angle, $\ker \mathbf{L}$ vs span($\mathbf{1}$, top 2 of $\mathbf{B}$)	0	0.0023
k-means on top 2 eigenvectors of $\mathbf{B}$: ARI vs components	1	1.000
ARI vs sectors		1.000

Table 3: Part I on the repaired Fig. 2 graph.

The partition prediction holds exactly: k-means on the top two eigenvectors of $\mathbf{B}$ recovers the components, which are the sectors. The spectral predictions hold to within the size of the degree deviation, and most of that deviation is a uniform shift. Every degree sits slightly above one, so the graph is closer to c-regular with $c = \bar{d} = 1.0047$ than to unit-degree. Reflecting about c instead of 1, as the regular row of Table 2 prescribes, cuts the largest spectral mismatch from 0.0059 to 0.0020 and the mean from 0.0047 to 0.0006 (Figure 1). This also explains why μ_1, μ_2 slightly exceed the ceiling of 1 in (a), and why the Fan identity misses by $-0.0093 \approx 2(1 - \bar{d})$.

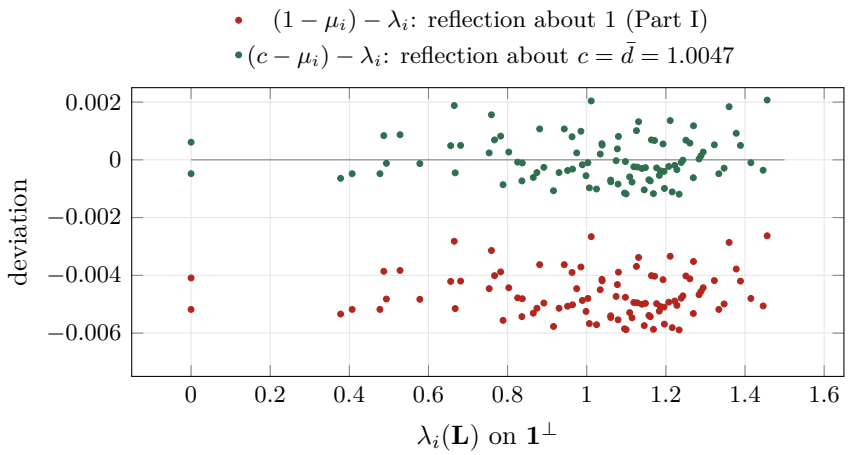


Figure 1: How far each of the 96 eigenvalue pairs on $\mathbf{1}^\perp$ is from Part I's prediction $\mu_i = 1 - \lambda_i$, on the repaired Fig. 2 graph (eigenvalues of $\mathbf{B}$ taken in decreasing order, of $\mathbf{L}$ in increasing order). The offset of about -0.005 is the mean degree's excess over one; once it is accounted for, what is left is the spread of the degrees around their mean.

7 Loosening the degree rule

To see the match fail as degrees spread, I reran Algorithm 1 with the same data, k, η and start, lowering only the degree penalty ρ (Table 4). The rank constraint keeps holding: every run returns three components and no isolated stock.

ρ	degree range	degree CV	components	rel. residual	μ_1, μ_2 of $\mathbf{B}$	sin	ARI vs comp.	μ_1, μ_2 of $\mathbf{B}_n$
100	[1.001, 1.007]	0.0014	52/28/17	0.017	1.005, 1.004	0.002	1.000	1, 1
30	[1.004, 1.023]	0.0047	52/28/17	0.056	1.017, 1.013	0.008	1.000	1, 1
10	[1.011, 1.064]	0.0129	52/28/17	0.152	1.048, 1.038	0.021	1.000	1, 1
3	[1.031, 1.184]	0.0344	52/28/17	0.392	1.139, 1.110	0.058	1.000	1, 1
1	[1.060, 1.421]	0.0700	49/31/17	0.730	1.318, 1.254	0.123	1.000	1, 1
0.3	[1.109, 1.881]	0.1265	52/28/17	1.098	1.647, 1.491	0.236	1.000	1, 1

Table 4: Algorithm 1 at weaker degree penalties. Relative residual is $\|\mathbf{B} - (\mathbf{I} - \mathbf{L} - \mathbf{1}\mathbf{1}^\top/p)\|_F / \|\mathbf{B}\|_F$; sin is the largest principal angle between $\ker \mathbf{L}$ and $\operatorname{span}(\mathbf{1}, \text{top 2 of } \mathbf{B})$; ARI compares k-means on the top two eigenvectors of $\mathbf{B}$ with the components. The eigenvalues of $\mathbf{B}_n$ equal 1 to six decimals in every row.

Three things stand out.

- **The algebra of Part I degrades steadily.** As the degree CV grows from 0.0014 to 0.126, the residual of (1) grows from 1.7% to more than the size of $\mathbf{B}$ itself, the top eigenvalues of $\mathbf{B}$ climb from 1.005 to 1.65, well past the ceiling of 1 in (a), and the subspaces drift apart.
- **The normalised identity does not degrade at all.** $\mathbf{B}_n$ keeps eigenvalue 1 with multiplicity exactly $k - 1 = 2$ at every ρ, as Section 5 predicts for any degrees.
- **The partition survives.** Even at $\rho = 0.3$, k-means on the eigenvectors of the combinatorial $\mathbf{B}$ returns the components exactly. With three components and no edges between them, the drift ($\sin \leq 0.24$) is not large enough to move any stock. I do not have a proof that this always happens; it is what these graphs show.

One run is out of line. At $\rho = 1$ the components are 49/31/17: three Industrials stocks, EFX, RSG and WM, join the Consumer Staples component (ARI against sectors 0.896). RSG and WM are waste-management companies, usually regarded as defensive, so the grouping is plausible, but I have not tested it further; the runs at $\rho = 3$ and $\rho = 0.3$ return the sectors exactly. It is a property of the learned graph, not of the Part I comparison, which still matches the components exactly in that row.

8 Scope of (c)

Part (c) is exact under unit degree, but the paper's crash indicator comes from its time-varying estimator, Problem (11), which has no degree constraint. As reported in the 18 October note, our 243 learned FAAMUNG graphs have mean degree 2.13 and degree CV 0.305, and $\lambda_2 \geq 1$ in 58% of windows. On those graphs $\lambda_{\max}(\mathbf{B}_t)$ and $\lambda_2(\mathbf{L}_t)$ are different quantities, and the sweep above shows how quickly the combinatorial match degrades once degrees spread (the FAAMUNG CV is more than twice the largest in Table 4). I have stated (c) for unit-degree graphs only. If a modularity indicator on the time-varying graphs is still of interest, the normalised version $\lambda_{\max}(\mathbf{B}_n) = \max\{0, 1 - \lambda_2(\mathbf{L}_{\text{sym}})\}$ holds for any degrees and could be the version to test; whether that is worth doing is your call.

9 Status

claim	status	evidence
(a) rank $\Leftrightarrow$ eigenvalue-1 multiplicity	proved; holds	Proposition 1; tests on 1 to 4 components; Fig. 2 graph gives $1.0052, 1.0041$ (degree tolerance)
(b) Fan = relaxed modularity	proved; holds under $\lambda_k \leq 1$	exact form with $\max\{0, \lambda_k - 1\}$; two counterexamples; Fig. 2 gap 0.0093
(c) $\mu_1 = \max\{0, 1 - \lambda_2\}$	proved; unit degree only	does not apply to Problem (11) graphs (Section 8)
regular / normalised / unequal	proved	Table 2; triangle-with-tails example; $\mathbf{B}_n$ exact in every sweep run
k-means on $\mathbf{B}$ recovers the components	reproduced	ARI 1.000 on Fig. 2 and at every ρ

Deviations recorded. The degree constraint is a quadratic penalty, so the learned degrees are 1.0012 to 1.0069 rather than 1, and every spectral comparison above carries that tolerance. The paper does not report η; $\eta = 300$ is ours. Algorithm 1 starts from the Problem (1) estimate, not the uniform graph (18 October note). The ρ sweep is ours, not the paper's.

Reproducing. `python report_numbers.py --part1` prints sections D to G of its output from the committed caches in a few seconds; `python degree_sweep.py 30 10 3 1 0.3` rebuilds the sweep; `tests/test_lemma.py` (25 tests) checks the lemma, the counterexamples and the classification on random graphs.

Next. With Part I settled, I would suggest Part II next, since it does not depend on the Part III repair: it starts with the invariance $\mathrm{tr}(\mathbf{LS}) = \mathrm{tr}(\mathbf{LJSJ})$ and a synthetic one-factor check of when unequal betas start to mislabel clusters. The order is your call.